FENGCHI LIU

Fuyang Normal University
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EDUCATION

Fuyang Normal University — Anhui, China
B.S. in Computer Science — GPA:3.74/5.0 (**12/68**) — Sep 2023 – Jul 2027

PUBLICATIONS

* indicates equal contribution

[CSIS-IAC 2026] **Adaptive Super-Twisting Sliding Mode Path-Tracking Control for Wheeled Agricultural Vehicles Subject to Sideslip Effects**
Fengchi Liu, Chen Ding, Wei Kang
International Annual Conference on Complex Systems and Intelligent Science(CSIS-IAC), 2026

[ICAIS&ISAS 2026] **Composite Adaptive Second-Order Sliding Mode Control for Path Tracking of Autonomous Agricultural Vehicles**
Fengchi Liu, Chen Ding, Qiankang Hou
Joint International Conference on Automation-Intelligence-Safety&International Symposium on Autonomous Systems, 2026

PREPRINTS AND MANUSCRIPTS

Design of Adaptive High-order Sliding Mode Controller Subject to Disturbances With Unknown State-Dependent Structures
Fengchi Liu, Chen Ding, Wei Kang, Zhangbao Xu, Qiankang Hou, Applied International Journal of Dynamics and Control, Minor revision

Reduced prescribed-time second-order sliding mode control for autonomous agricultural vehicle path tracking with deferred asymmetric performance constraints
Fengchi Liu, Chen Ding, Wei Kang, Zhangbao Xu, Qiankang Hou, Applied Nonlinear Dynamics, With editor

RESEARCH EXPERIENCE

- *Fuyang Normal University* — *Anhui, China*

  Research assistant, supervised by Prof. Chen Ding. — Apr 2024 – Now

  Presided over 2025 national-level College Students' Innovative Training Plan Program
  - Developed a MATLAB simulation model for an unmanned vehicle path-tracking control system;
  - Designed and implemented a disturbance-rejection control algorithm;
  - Verified its effectiveness and superior performance through simulation studies.

AWARDS AND HONORS

- Outstanding Student Scholarship, Fuyang Normal University — 2025
- Merit Student, Fuyang Normal University — 2025
- Provincial Second Prize, Lanqiao Cup National Software and Information Technology Professionals Competition (C/C++ Track) — 2025
- Provincial Third Prize, ROBOCOM Robot Developers Competition — 2025
- Successful Participant, Mathematical Contest in Modeling (MCM) — 2025
- Honorable Mention, MathorCup Mathematical Modeling International Competition — 2024
- Outstanding Award, MathorCup Mathematical Modeling Online Challenge — 2024
- University Gold Award, National College Students' Innovation, Creativity and Entrepreneurship Challenge — 2024
- College First Prize, "Challenge Cup" College Student Entrepreneurship Plan Competition — 2024